Second-order Sobolev regularity of three-dimensional p-harmonic functions

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Abstract

For every finite $p \geq 5$, a weak p-harmonic function on an arbitrary open subset of $\mathbb{R}^3$ belongs to $W^{2,2}_{\mathrm{loc}}$. The conclusion concerns the full unweighted distributional Hessian and includes neighborhoods of critical points without assumptions on the critical set. We prove an estimate for smooth regularized solutions in which the Hessian energy is divided by the actual gradient norm. Gaussian curvature of the level surfaces supplies the component that is lost in the usual pointwise coercivity estimate. The equation excludes closed level components in a ball, so Gauss–Bonnet and the total curvature of the boundary curves control the integrated correction. A radial cutoff argument makes the estimate uniform in the regularization parameter, and smooth approximation identifies every second distributional derivative.

1 Introduction

The p-Laplace equation

$$\operatorname{div}\bigl(|Du|^{p-2}Du\bigr) = 0 \tag{1}$$

is a basic model of degenerate ellipticity. Its weak solutions are p-harmonic functions. Local Hölder continuity of the gradient is available for every $1 < p < \infty$, but for $p > 2$ the coefficient in (1) degenerates where the gradient vanishes [1, Section 4]. Consequently, this first-order regularity does not by itself identify a square-integrable Hessian across the critical set.

Second-order estimates often control a nonlinear function of the gradient. Sarsa proves local $W^{1,2}$ regularity of $|Du|^{(p-2+\sigma)/2}Du$ when $\sigma > -1 - (p-1)/(n-1)$ [2, Theorem 1.1]. Setting $\sigma = 2 - p$ removes the weight only in the range $p < 3 + 2/(n-2)$ for $n > 2$. The same range appears in the classical unweighted $W^{2,2}_{\mathrm{loc}}$ result discussed by Lindqvist [1, Section 11, p. 93, Second Derivatives] and in the global Cordes-based theory of Haarala and Sarsa [3]. In three dimensions its upper endpoint is 5. A weighted derivative estimate does not settle what happens at a zero of Du when the weight vanishes, and the pointwise coercivity underlying that argument does not remain positive beyond the endpoint.

We prove the unweighted conclusion in the remaining range. We use the usual distributional definition: $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$ is a weak solution if

$$\int_\Omega |Du|^{p-2}Du \cdot D\varphi \, dx = 0 \qquad \bigl(\varphi \in C_c^\infty(\Omega)\bigr). \tag{2}$$

Theorem 1.1. *Let $5 \leq p < \infty$, let $\Omega \subset \mathbb{R}^3$ be open, and let $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$ be real valued and satisfy (2). Then $u \in W^{2,2}_{\mathrm{loc}}(\Omega)$. In particular, each of the nine distributional derivatives D_jD_iu, $1 \leq i, j \leq 3$, belongs to $L^2_{\mathrm{loc}}(\Omega)$.*

Here $W^{k,r}_{\mathrm{loc}}$ means that the function and all its distributional derivatives of order at most k are locally in L^r. Theorem 1.1 imposes no sign or symmetry condition and no restriction on the geometry or measure of the critical set. It gives an affirmative answer to the three-dimensional second-derivative question stated in Section 11 of Lindqvist's notes, including $p = 5$ and every larger finite exponent.

The missing normal component of the Hessian can be recovered by adding signed Gaussian curvature of the level surfaces to a divergence expression. This correction need not be positive at each point. Its integral is nevertheless bounded by quantities on the boundary of a ball: the equation rules out a closed regular level component, and Gauss–Bonnet reduces its signed curvature integral to the total curvature of the boundary curves. Fenchel's inequality then controls the number of those curves.

The resulting estimate divides the squared Hessian by $|Dv|$ for a smooth regularized solution v. At a critical point this quotient requires care even though v is smooth. A nonnegative norm identity and Fatou's lemma establish its local integrability and the favorable distributional inequality before curvature is integrated. Radial cutoffs and an iteration remove the preliminary dependence on the regularization parameter. The known uniform gradient bound then converts this estimate into an unweighted L^2 bound.

All constants below may depend on the fixed finite exponent p; we make no assertion uniform as $p \to \infty$. The argument uses two-dimensional level surfaces and does not assert the corresponding result in higher spatial dimensions. Section 2 treats critical points of smooth regularizations. Section 3 supplies the curvature identity and its boundary estimate, and Section 4 proves the uniform estimate and the distributional conclusion. Appendix A gives the smooth critical-value and coarea facts used in the geometric step. All vector norms are Euclidean and matrix norms are Hilbert–Schmidt; balls in an argument have a common center.

2 A norm identity through the critical set

Fix $5 \le p < \infty$, $\varepsilon > 0$, and a smooth real solution v on an open set $U \subset \mathbb{R}^3$ of

$$\operatorname{div}\big((\varepsilon + |Dv|^2)^{(p-2)/2} Dv\big) = 0. \tag{3}$$

The approximation to a general weak solution will be taken only after the estimates for this fixed v have been proved. Write

$$q = |Dv|, \qquad H = D^2 v, \qquad Z = \{q = 0\}.$$

On $U \setminus Z$, let $e = Dv/q$, and split the Hessian into its normal and tangential blocks:

$$\begin{gathered} P = I - e \otimes e, \qquad a = e^T H e, \qquad w = PHe, \\ B = PHP\big|_{e^\perp}, \qquad B_0 = B - \tfrac{1}{2}(\operatorname{tr} B) I_{e^\perp}, \qquad s = 1 + (p-2)\frac{q^2}{q^2+\varepsilon}. \end{gathered} \tag{4}$$

The block B is a symmetric operator on the two-dimensional tangent plane. Expansion of (3) gives

$$\operatorname{tr} H + (s-1)a = 0, \qquad \operatorname{tr} B = -sa, \qquad 1 \le s \le p-1. \tag{5}$$

In an orthonormal frame with first vector e,

$$|H|^2 = (1 + s^2/2)a^2 + 2|w|^2 + |B_0|^2, \qquad T := |H|^2 - |He|^2 = |w|^2 + |B|^2. \tag{6}$$

Since $s^2 a^2 \le 2|B|^2$ and $s \ge 1$, these identities imply

$$|H|^2 \le 3T. \tag{7}$$

Lemma 2.1. *Define*

$$E = \frac{|H|^2}{q} \quad \text{on } U \setminus Z, \qquad E = 0 \quad \text{on } Z.$$

Then $E \in L^1_{\mathrm{loc}}(U)$. The weak gradient of q is He, with $e = 0$ on Z. The vector field

$$G = (\Delta v)e, \qquad S = Dq - G$$

has a C^1 extension for G and a locally bounded realization for S, both zero on Z. If

$$f = \frac{|w|^2 + |B_0|^2 - \frac{1}{2}s(s-2)a^2}{q} \quad \text{on } U \setminus Z, \qquad f = 0 \quad \text{on } Z, \tag{8}$$

then $f \in L^1_{\mathrm{loc}}(U)$ and

$$\int_U \psi f \, dx \le -\int_U S \cdot D\psi \, dx \qquad (0 \le \psi \in C_c^\infty(U)). \tag{9}$$

Moreover, $|S| \le (1+\sqrt{3})|H|$ almost everywhere, and

$$|H| = \sqrt{E}\,\sqrt{q} \qquad \text{almost everywhere on } U. \tag{10}$$

Proof. For $\delta > 0$ set $q_\delta = (q^2 + \delta^2)^{1/2}$. Differentiation gives

$$Dq_\delta = \frac{HDv}{q_\delta}, \qquad \Delta q_\delta = \underbrace{\frac{|H|^2}{q_\delta} - \frac{|HDv|^2}{q_\delta^3}}_{T_\delta} + \frac{Dv}{q_\delta} \cdot D\Delta v.$$

Here $T_\delta \ge 0$ because $|HDv| \le |H|q$. On compact subsets, $q_\delta \to q$ uniformly and $Dq_\delta \to He$ pointwise, including zero on Z; also $|Dq_\delta| \le |H|$. Dominated convergence in the test identity for a first derivative identifies the weak gradient $Dq = He$. For a nonnegative compactly supported smooth ψ, integration by parts yields

$$\int \psi T_\delta \, dx = \int q_\delta \Delta\psi \, dx - \int \psi \frac{Dv}{q_\delta} \cdot D\Delta v \, dx.$$

The right side has a finite limit: $D\Delta v$ is bounded on the support and $|Dv/q_\delta| \le 1$. Fatou's lemma, applied on $U \setminus Z$, therefore gives

$$\int_{U\setminus Z} \psi \frac{T}{q} \, dx \le -\int Dq \cdot D\psi \, dx - \int \psi e \cdot D\Delta v \, dx. \tag{11}$$

Choose ψ equal to one on a prescribed compact set. Equations (7) and (11) imply $E \in L^1_{\mathrm{loc}}$. This is a finiteness statement for the fixed smooth solution; its present bound may depend on ε and on third derivatives.

We next justify integration by parts for G at Z. Equation (5) implies

$$G = -(p-2)\frac{(Dv^T HDv)Dv}{q(q^2+\varepsilon)} \qquad (q > 0). \tag{12}$$

For a symmetric matrix A, the map $F(z, A) = (z^T Az)z/|z|$, assigned zero at $z = 0$, is C^1. Indeed, off zero

$$|F| \le |A||z|^2, \qquad |D_z F| \le C|A||z|, \qquad |D_A F| \le C|z|^2.$$

The first bound shows that the derivative at zero is zero, and the other bounds give its continuity there. Composing with Dv, H and dividing by the positive smooth function $q^2 + \varepsilon$ in (12) proves that G is C^1 and $G = DG = 0$ on Z.

Off Z, differentiation of e gives $\operatorname{div} e = (\Delta v - a)/q$, so the classical divergence of S is

$$\frac{T - (\Delta v)^2 + (\Delta v)a}{q}.$$

Using $\Delta v = (1 - s)a$ and (6) gives exactly (8). Equivalently, off Z,

$$f = T/q + e \cdot D\Delta v - \operatorname{div} G.$$

All three terms are locally integrable. The latter two vanish on Z, and T/q is extended by zero there. Adding those two terms to (11) and integrating G by parts proves (9). In particular, the passage through the critical set preserves the required direction of the inequality. Since $|Dq| \le |H|$ and $|\Delta v| \le \sqrt{3}|H|$, the stated bound for S follows.

Finally $H = 0$ almost everywhere on Z. For each smooth component v_i of Dv, the part of $\{v_i = 0\}$ where $Dv_i \neq 0$ is locally a smooth hypersurface by the implicit function theorem. A graph has three-dimensional measure zero by Fubini, and countably many such graphs cover that part of the zero set. Apply this to all three components v_i. At almost every point of Z all the rows of H vanish, proving (10). The convergence and integration facts used here are the usual Fatou, dominated convergence and Fubini theorems [6, Sections 1.4 and 1.7]; the smooth calculus and implicit graph facts can be found in [8, Sections 4.3 and 5.3] and [9, Sections 8.3, 8.5 and 8.6]. □

3 Curvature of level surfaces

3.1 The coercive identity

At $q > 0$, the second fundamental form of the level surface of v is B/q, up to the sign convention for the normal. Its Gaussian curvature is therefore $K = \det B/q^2$. Introduce the signed density

$$\mathcal{K} = Kq = \frac{\det B}{q} = \frac{s^2a^2/4 - |B_0|^2/2}{q} \quad (q > 0), \qquad \mathcal{K} = 0 \quad (q = 0). \tag{13}$$

The determinant of a symmetric two-dimensional matrix satisfies $|\det B| \le |B|^2/2$, whence $|\mathcal{K}| \le E/2$. Lemma 2.1 has thus established absolute integrability before we integrate signed curvature.

Let

$$\ell = p - 1, \qquad c_p = \ell + \ell^{-1}, \qquad d_p = 2c_p - 2. \tag{14}$$

Substitution of (6), (8), and (13) gives the identity

$$c_p f + d_p\mathcal{K} - E = \frac{(c_p - 2)|w|^2 + (sc_p - s^2 - 1)a^2}{q} \qquad (q > 0). \tag{15}$$

The coefficient of $|B_0|^2$ cancels. The remaining normal coefficient factors as

$$sc_p - s^2 - 1 = \frac{(\ell - s)(s\ell - 1)}{\ell} \ge 0 \qquad (1 \le s \le \ell),$$

and $c_p - 2 = (\ell - 1)^2/\ell \ge 0$. Consequently

$$E \le c_p f + d_p\mathcal{K} \qquad \text{almost everywhere on } U. \tag{16}$$

There is no separate positivity assertion for f or for K.

3.2 An integrated boundary estimate

Lemma 3.1. *For every closed ball $\overline{B_r} \subset U$,*

$$\int_{B_r} \mathcal{K}\, dx \le 2 \int_{\partial B_r} \left(|H| + \frac{2q}{r} \right) dA. \tag{17}$$

Proof. Translate the center to zero. By Lemma A.1, almost every t is a regular value both of v near the ball and of $v|_{\partial B_r}$. For such t, $\Sigma_t = \{v = t\} \cap \overline{B_r}$ is a compact smooth orientable surface with smooth boundary, possibly empty. Its orientation comes from Dv/q. It has finitely many components, as does its boundary: connected coordinate neighborhoods have a finite subcover of a compact manifold and each belongs to one component.

Every nonempty component has boundary. Otherwise a connected closed embedded component M lies in B_r. It separates a bounded region D from the unbounded region [4, B19]. Every point outside the ball can be joined to infinity without meeting M, so $D \subset B_r$, and $v = t$ on $\partial D = M$. To use the equation without importing a boundary regularity argument, freeze the coefficient field

$$A(x) = I + (p-2)\frac{Dv \otimes Dv}{q^2 + \varepsilon}, \qquad Lz = \operatorname{tr}(A(x) D^2 z).$$

Then $Lv = 0$ and $I \le A \le (p-1)I$. For $\rho > 0$, $L(v + \rho e^{x_1}) = \rho A_{11} e^{x_1} > 0$. At an interior maximum the Hessian is negative semidefinite, which would give $L(v + \rho e^{x_1}) \le 0$. Thus its maximum on $\overline{D}$ is attained on ∂D. Letting $\rho \downarrow 0$ gives $v \le t$ in D. Applying the minimum argument to $v - \rho e^{x_1}$ gives $v \ge t$. Hence $v = t$ in D, and continuity of Dv gives $Dv = 0$ on M, contradicting regularity of the level.

Let b be the total number of boundary curves. A connected orientable compact surface of genus γ_i with b_i boundary circles has Euler characteristic $2 - 2\gamma_i - b_i$ [4, B17]. Here $b_i \ge 1$, so $\chi(\Sigma_t) \le \sum_i b_i = b$. Gauss–Bonnet for a smooth boundary [4, B16–B18] gives

$$\begin{aligned} \int_{\Sigma_t} K\, dA &= 2\pi\chi(\Sigma_t) - \sum_{\Gamma} \int_{\Gamma} k_g^{\Sigma_t}\, ds \\ &\le 2\pi b + \sum_{\Gamma} \int_{\Gamma} |k_g^{\Sigma_t}|\, ds \le 2 \sum_{\Gamma} \int_{\Gamma} \kappa\, ds. \end{aligned} \tag{18}$$

Here $k_g^{\Sigma_t}$ is signed geodesic curvature with the Gauss–Bonnet orientation, and κ is nonnegative Euclidean curvature of a boundary curve. The last inequality uses $|k_g^{\Sigma_t}| \le \kappa$ and Fenchel's inequality $2\pi \le \int_\Gamma \kappa\, ds$ separately for each closed regular curve [5, Theorem 2.4 and Proposition 3.1].

We bound this last curvature on the sphere. Put $n = x/r$ and $g = |D_{\partial B_r} v|$. On $\{g > 0\}$ an arclength parametrization Γ of a spherical level curve has unit tangent T and

$$\Gamma'' = \eta \nu - \frac{n}{r}, \qquad \nu = \frac{D_{\partial B_r} v}{g},$$

for a signed scalar η. Indeed, differentiation of $n \cdot T = 0$ gives the normal component $-n/r$, while the tangential component is perpendicular to T. Differentiating $v(\Gamma) = t$ twice gives $0 = H(T,T) + \eta g - D_n v / r$. It follows that

$$\kappa g \le |\eta| g + \frac{g}{r} \le |H| + \frac{|D_n v| + g}{r} \le |H| + \frac{2q}{r}. \tag{19}$$

The volume coarea identity in Lemma A.2 gives

$$\int_{B_r} \mathcal{K}\, dx = \int_{\mathbb{R}} \int_{\Sigma_t} K\, dA\, dt.$$

It applies to this signed integrand because $|\mathcal{K}| \le E/2$ is integrable. Surface coarea on ∂B_r gives

$$\int_{\mathbb{R}} \sum_{\Gamma} \int_{\Gamma} \kappa \, ds \, dt = \int_{\partial B_r} \kappa g \, dA,$$

with $\kappa g = 0$ on $\{g = 0\}$. This nonnegative integral is finite by (19). Integrate (18) in t and use (19) to obtain (17). The critical-value assertion concerns the fixed smooth v, and does not make any assertion about critical values of the limiting weak solution. □

4 Uniform energy and distributional second derivatives

Proposition 4.1. *For each finite $p \ge 5$ there is $L_p < \infty$, depending only on p, such that every smooth solution of (3) on a neighborhood of $\overline{B_{2R}}$ satisfies*

$$\int_{B_R} \frac{|D^2 v|^2}{|Dv|} \mathbf{1}_{\{|Dv|>0\}} \, dx \le \frac{L_p}{R^2} \int_{B_{2R}} |Dv| \, dx. \tag{20}$$

The constant is independent of $\varepsilon > 0$ and of v.

Proof. Retain $q = |Dv|$, $H = D^2 v$, and the zero-extended densities $E, \mathcal{K}$ from Sections 2 and 3. Set $F(r) = \int_{B_r} E \, dx$ and $Q = \int_{B_{2R}} q \, dx$. These are finite for this fixed smooth solution. For $R \le a < b \le 2R$ put $h = b - a$, and choose a smooth decreasing function η, equal to one for $r \le a$ and zero for $r \ge b$, with $0 \le \eta \le 1$ and $|\eta'| \le 2/h$. Such a function is obtained by integrating a smooth nonnegative unit-mass function supported in $(0, 1)$ and bounded by 2, then scaling the interval. The radial function $\psi(x) = \eta(|x|)$ is a smooth compactly supported test, constant near the center. Equations (9) and (16) yield

$$F(a) \le \int \psi E \, dx \le -c_p \int S \cdot D\psi \, dx + d_p \int \psi \mathcal{K} \, dx. \tag{21}$$

The identity $\psi(x) = \int_{|x|}^{b} -\eta'(r) \, dr$ and Fubini give

$$\int \psi \mathcal{K} \, dx = \int_a^b -\eta'(r) \int_{B_r} \mathcal{K} \, dx \, dr.$$

The interchange is justified by absolute integrability of $\mathcal{K}$, despite its sign. Apply Lemma 3.1 under this nonnegative radial weight, and use polar integration on the shell $B_b \setminus B_a$. The result is

$$\int \psi \mathcal{K} \, dx \le \frac{4}{h} \int_{B_b \setminus B_a} |H| \, dx + \frac{8}{ah} \int_{B_b \setminus B_a} q \, dx. \tag{22}$$

Polar integration here is the smooth coarea identity for $|x|$ on the shell. Also, Lemma 2.1 gives

$$-\int S \cdot D\psi \, dx \le \frac{2(1+\sqrt{3})}{h} \int_{B_b \setminus B_a} |H| \, dx.$$

Put

$$A_p = 2c_p(1+\sqrt{3}) + 4d_p, \qquad B_p = 8d_p.$$

By (10), Cauchy–Schwarz, and (21)–(22),

$$\begin{aligned} F(a) &\le \frac{A_p}{h} \sqrt{F(b) Q} + \frac{B_p}{ah} Q \\ &\le \vartheta F(b) + \frac{A_p^2/(4\vartheta) + B_p}{h^2} Q \qquad (0 < \vartheta < 1). \end{aligned} \tag{23}$$

The second line uses $xy \le \vartheta x^2 + y^2/(4\vartheta)$ and $h \le R \le a$.

Choose $\vartheta = 1/8$ and write $L = A_p^2/(4\vartheta) + B_p$. With $r_j = 2R - R2^{-j}$, iteration of (23) gives

$$F(R) \le \vartheta^m F(r_m) + \frac{4LQ}{R^2} \sum_{j=0}^{m-1} (4\vartheta)^j.$$

For the fixed ε and v, $F(r_m) \le F(2R) < \infty$. Thus the terminal term tends to zero as $m \to \infty$. Since $4\vartheta = 1/2$, (20) follows with $L_p = 8L$. This iteration is completed before taking any limit in ε; no uniform bound on the preliminary quantity $F(2R)$ was used. □

Proof of Theorem 1.1. Fix an arbitrary ball $\overline{B_{4R}} \subset \Omega$. The local $C^{1,\alpha}$ representative of u is available by [1, Section 4]. Use the smooth Dirichlet approximation on $U = B_{4R}$ described in [2, Section 3, (3.2)–(3.4)]: for a sequence $\varepsilon_k \downarrow 0$, there are $v_k \in C^\infty(U)$ solving (3) with $\varepsilon = \varepsilon_k$ and boundary data u, such that the interior $C^{1,\alpha}$ norms are uniformly bounded and $Dv_k \to Du$ locally uniformly. These approximation statements hold for every finite $p > 1$; they do not require a second-order bound on u. In the present application, u is bounded on $\overline{U}$ by its local C^1 regularity, the ball is smooth, and $\overline{B_{2R}} \subset U$ has positive distance from its boundary. Consequently

$$M := \sup_k \sup_{B_{2R}} |Dv_k| < \infty.$$

Lemma 2.1 says that $D^2 v_k = 0$ almost everywhere where $Dv_k = 0$. Proposition 4.1 therefore gives the full unweighted bound

$$\int_{B_R} |D^2 v_k|^2 \, dx \le \frac{L_p M}{R^2} \int_{B_{2R}} |Dv_k| \, dx \le \frac{L_p M^2 |B_{2R}|}{R^2} =: C_0^2. \tag{24}$$

This includes the case $M = 0$. For every $i, j \in \{1, 2, 3\}$ and $\varphi \in C_c^\infty(B_R)$, local uniform gradient convergence and integration by parts imply

$$-\int_{B_R} D_i u \, D_j \varphi \, dx = \lim_{k \to \infty} \int_{B_R} D_j D_i v_k \, \varphi \, dx.$$

By (24), its absolute value is at most $C_0 \|\varphi\|_{L^2(B_R)}$. Thus $D_j D_i u$, initially a distribution, defines a bounded linear functional on $C_c^\infty(B_R)$ with the L^2 norm.

For completeness, density on this ball follows from the usual $C_c^\infty(\mathbb{R}^3)$ density in $L^2(\mathbb{R}^3)$ and smooth cutoffs [7, Exercise 1.13.1(iii)–(iv)]. Extend $h \in L^2(B_R)$ by zero and take cutoffs $0 \le \eta_m \le 1$ supported in B_R with $\eta_m \to 1$ pointwise there. Dominated convergence gives $\eta_m h \to h$ in L^2. For fixed m, approximate the zero extension of h by smooth compactly supported functions on $\mathbb{R}^3$ and multiply by η_m. A diagonal choice proves density of $C_c^\infty(B_R)$ in $L^2(B_R)$. The functional extends uniquely to that real Hilbert space. Riesz representation [7, Theorem 1.4.13] supplies $h_{ji} \in L^2(B_R)$ such that

$$-\int_{B_R} D_i u \, D_j \varphi \, dx = \int_{B_R} h_{ji} \varphi \, dx \qquad (\varphi \in C_c^\infty(B_R)).$$

This is precisely the distributional identity $D_j D_i u = h_{ji}$. It treats all nine entries without any condition on the limiting critical set. Because $p \ge 5 > 2$ and B_R has finite measure, $u, Du \in L^2(B_R)$ by Hölder's inequality. Hence $u \in W^{2,2}(B_R)$. Every point of Ω admits such balls. Distributional uniqueness identifies the derivatives on overlaps, and finite covers of compact subsets give their local square integrability. □

A Smooth critical values and coarea

The geometric argument needs these facts only for a fixed smooth regularization and its restriction to a sphere. The following proofs also specify the measures in the two coarea identities.

Lemma A.1. *If f is smooth on an open subset of $\mathbb{R}^m$, $1 \le m \le 3$, then the set of values at points where $Df = 0$ has one-dimensional Lebesgue measure zero. The same holds on a smooth m-dimensional manifold with a countable atlas.*

Proof. Induct on m. Let A_j be the set where all derivatives of orders $1, \dots, j$ vanish. On a compact subset of the domain, divide a bounding rectangle into cubes of side h. For a cube meeting A_m, choose x in that intersection. Taylor's theorem applied to $\tau \mapsto f(x+\tau(y-x))$ gives $|f(y)-f(x)| \le Ch^{m+1}$ for y in that cube: the derivatives through order m vanish at x, and the $(m+1)$st derivatives are bounded on a fixed neighborhood. There are at most $C'h^{-m}$ cubes. The outer measure of the image of A_m on this compact subset is therefore at most $C''h$, and hence zero. A compact exhaustion removes the compactness restriction.

For $1 \le j < m$ and $x \in A_j \setminus A_{j+1}$, some derivative $D^\beta f$ of order j has a nonzero first derivative at x. Its zero set is a smooth $(m-1)$-dimensional hypersurface near x and contains A_j there. Since $Df = 0$ on A_j, these points are critical for the restriction of f to that hypersurface. The induction hypothesis, in its charts, makes their image null. Countably many such neighborhoods cover each stratum. Their union with A_m is A_1. For $m = 1$ only the Taylor argument is needed. Countably many charts also establish the manifold statement. Taylor's theorem and the implicit function theorem used here are [8, Theorem 4.3.2] and [9, Theorem 8.5.6]. □

Lemma A.2. *Let f be smooth on an open subset V of $\mathbb{R}^3$. For a Borel function h on $\{|Df| > 0\}$,*

$$\int_{\{|Df|>0\}} h|Df|\,dx = \int_{\mathbb{R}} \int_{\{f=t\}\cap\{|Df|>0\}} h\,dA\,dt \tag{25}$$

if $h \ge 0$, or if the integral of $|h||Df|$ is finite. For a smooth surface N and $g = |D_N f|$, the corresponding identity is

$$\int_{\{g>0\}\subset N} hg\,dA = \int_{\mathbb{R}} \int_{\{f=t\}\cap\{g>0\}} h\,ds\,dt \tag{26}$$

under the analogous condition.

Proof. In a coordinate rectangle with $f_1 \ne 0$, solve $f(x_1, y) = t$ for x_1. One-variable substitution gives $dx_1 = dt/|f_1|$, and the level graph has area element $|Df|\,dy/|f_1|$. These equalities and Fubini give (25) on the rectangle. To justify this substitution for Borel integrands, first apply the elementary substitution theorem to intervals [8, Theorem 5.3.5]. The two finite measures agree on their Boolean algebra of finite interval unions and therefore on Borel sets by the monotone-class lemma [6, Lemma 1.7.14]. Exhaustion, simple functions and monotone convergence give the nonnegative formula. Positive and negative parts give the absolutely integrable formula.

For the surface identity, write a chart $X(y_1, y_2)$ with metric $G_{ij} = X_i \cdot X_j$. On a chart where $f_1 \ne 0$, the speed of the level curve parametrized by y_2 is

$$L = \left| X_2 - \frac{f_2}{f_1} X_1 \right|, \qquad g^2 = \frac{G_{22}f_1^2 - 2G_{12}f_1f_2 + G_{11}f_2^2}{\det G}.$$

Thus $L = g\sqrt{\det G}/|f_1|$, while substitution of t in the surface area element gives $\sqrt{\det G}\,dt\,dy_2/|f_1|$. This proves (26) in the chart by the same substitution argument. Cover each noncritical set with countably many such charts, partition it into disjoint Borel subsets of the charts and sum using Tonelli. Absolute integrability permits the signed sum. Lemma A.1 allows the t integrals to be restricted to simultaneous regular values. □

AI-use statement

AI assistance was used in developing and checking the argument, locating and checking sources, and preparing the manuscript. The submission is offered for independent mathematical review.

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